



United International University
School of Science and Engineering
Course Name: Coordinate Geometry & Vector Analysis
Course Code: MATH-2201, **Practice Sheet (for Mid Exam)**

Chapter: 11.1

1. Find the distance from the point $A(1, -1, 1)$ to

(a) $B(2, 0, -3)$

(b) xy - plane.

(c) z -axis.

2. Identify and hence sketch

(a) $y^2 + z^2 = 4$

(b) $x^2 + y^2 + z^2 + 4x - 6y + 8z - 7 = 0$

Chapter: 11.2

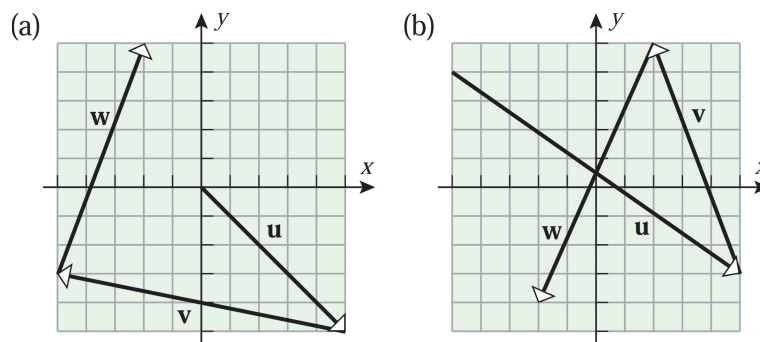
3. Find the vector $\mathbf{u} = \overrightarrow{P_1P_2}$, where $P_1 = (2, -1, 3)$ and $P_2 = (1, 2, -1)$.

4. Given that $\mathbf{u} = \mathbf{i} - \mathbf{j}$ is a vector.

(a) Find the unit vectors parallel to $\mathbf{u}$.

(b) Find the angle that the vector $\mathbf{u}$ makes with positive x - axis.

5. In each part, sketch $\mathbf{u} + \mathbf{v} + \mathbf{w}$ and express it in component form.



6. Find the vector $\mathbf{u}$ of length 3 which

(a) makes an angle of $\frac{\pi}{6}$ with x -axis.

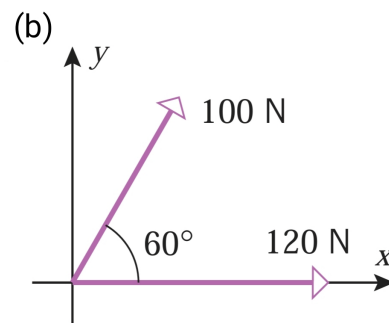
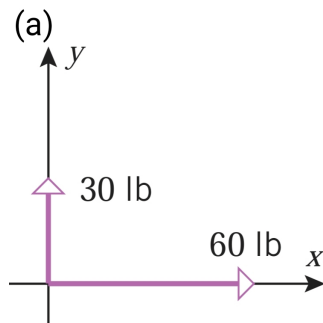
(b) is parallel to the vector $\mathbf{v} = 2\mathbf{i} - \mathbf{j} - 3\mathbf{k}$.

7. Given that $\mathbf{u} = 3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$ and $\mathbf{v} = -\mathbf{i} + 5\mathbf{k}$ are two vectors. Find the components of the following vectors.

(a) $\mathbf{u} - 3\mathbf{v}$

(b) $2\mathbf{v} - 5\mathbf{u}$

8. In each part, find and sketch the resultant force. Also, calculate the angle that the resultant force makes with x -axis.



Chapter: 11.3

9. Consider $\mathbf{u} = \langle 1, -2, 3 \rangle$, $\mathbf{v} = \langle -1, 1, 0 \rangle$ and $\mathbf{w} = \langle 3, 0, -2 \rangle$.

(a) Find the angle between $\mathbf{v}$ and $\mathbf{w}$.

(b) Find the direction cosines of $\mathbf{u}$.

(c) Find the vector component (orthogonal projection) of $\mathbf{v}$ along $\mathbf{u}$ and orthogonal to $\mathbf{u}$.

(d) Check whether the vectors $\mathbf{u}$, $\mathbf{w}$ are perpendicular to each other or not.

10. Consider $\mathbf{v} = \langle -1, 2 \rangle$, $\mathbf{e}_1 = \left\langle \frac{-1}{2}, \frac{\sqrt{3}}{2} \right\rangle$ and $\mathbf{e}_2 = \left\langle \frac{1}{2}, \frac{\sqrt{3}}{2} \right\rangle$.

(a) Find the scalar components of $\mathbf{v}$ along $\mathbf{e}_1$ and $\mathbf{e}_2$.

(b) Find the vector components of $\mathbf{v}$ along $\mathbf{e}_1$ and $\mathbf{e}_2$.

Chapter: 11.4

11. Find the area of the triangle with vertices P , Q , and R .
- (a) $P(1, 5, -2)$, $Q(0, 0, 0)$, $R(3, 5, 1)$
 - (b) $P(2, 0, -3)$, $Q(1, 4, 5)$, $R(7, 2, 9)$
12. Find the unit vectors perpendicular to both $\mathbf{u}$, $\mathbf{v}$. Also find scalar triple product for each of the following parts.
- (a) $\mathbf{u} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$, $\mathbf{v} = 4\mathbf{i} + \mathbf{j} - 3\mathbf{k}$, $\mathbf{w} = \mathbf{j} + 5\mathbf{k}$
 - (b) $\mathbf{u} = \langle 1, -2, 2 \rangle$, $\mathbf{v} = \langle 0, 3, 2 \rangle$, $\mathbf{w} = \langle -4, 1, -3 \rangle$
 - (c) $\mathbf{u} = \langle 2, 1, 0 \rangle$, $\mathbf{v} = \langle 1, -3, 1 \rangle$, $\mathbf{w} = \langle 4, 0, 1 \rangle$
 - (d) $\mathbf{u} = \mathbf{i}$, $\mathbf{v} = \mathbf{i} + \mathbf{j}$, $\mathbf{w} = \mathbf{i} + \mathbf{j} + \mathbf{k}$
13. Use a scalar triple product to find the volume of the parallelepiped that has $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ as adjacent edges.
- (a) $\mathbf{u} = \langle 2, -6, 2 \rangle$, $\mathbf{v} = \langle 0, 4, -2 \rangle$, $\mathbf{w} = \langle 2, 2, -4 \rangle$
 - (b) $\mathbf{u} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$, $\mathbf{v} = 4\mathbf{i} + 5\mathbf{j} + \mathbf{k}$, $\mathbf{w} = \mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$
14. In each part, determine whether the vectors lie in the same plane.
- (a) $\mathbf{u} = \langle 1, -2, 1 \rangle$, $\mathbf{v} = \langle 3, 0, -2 \rangle$, $\mathbf{w} = \langle 5, -4, 0 \rangle$
 - (b) $\mathbf{u} = 5\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{v} = 4\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{w} = \mathbf{i} - \mathbf{j}$
 - (c) $\mathbf{u} = \langle 4, -8, 1 \rangle$, $\mathbf{v} = \langle 2, 1, -2 \rangle$, $\mathbf{w} = \langle 3, -4, 12 \rangle$
15. Suppose that $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = -11$. Find:
- (a) $\mathbf{u} \cdot (\mathbf{w} \times \mathbf{v})$
 - (b) $(\mathbf{v} \times \mathbf{u}) \cdot \mathbf{w}$
 - (c) $\mathbf{w} \cdot (\mathbf{u} \times \mathbf{v})$
 - (d) $\mathbf{v} \cdot (\mathbf{u} \times \mathbf{w})$
 - (e) $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$

Chapter: 11.5

16. Find parametric equations of the line that satisfies the stated conditions.
- (a) The line through $(-5, 2)$ that is parallel to $2\mathbf{i} - 3\mathbf{j}$.

(b) The line through $(0, 3)$ that is parallel to the line

$$x = -5 + t, \quad y = 1 - 2t.$$

(c) The line through $(-1, 2, 4)$ that is parallel to $3\mathbf{i} - 4\mathbf{j} + \mathbf{k}$.

(d) The line through $(2, -1, 5)$ that is parallel to $\langle -1, 2, 7 \rangle$.

(e) The line through $(-2, 0, 5)$ that is parallel to the line

$$x = 1 + 2t, \quad y = 4 - t, \quad z = 6 + 2t.$$

(f) The line through the origin that is parallel to the line

$$x = t, \quad y = -1 + t, \quad z = 2.$$

17. Where does the line

$$x = 1 + 3t, \quad y = 2 - t$$

intersect

(a) the x -axis

(b) the y -axis

18. Find the intersections of each of the following lines with the xy -plane, the xz -plane, and the yz -plane.

$$x = -2, \quad y = 4 + 2t, \quad z = -3 + t.$$

$$x = -1 + 2t, \quad y = 3 + t, \quad z = 4 - t.$$

19. Show that the lines L_1 and L_2 intersect, and find their point of intersection.

(a)

$$L_1 : x = 2 + t, \quad y = 2 + 3t, \quad z = 3 + t$$

$$L_2 : x = 2 + t, \quad y = 3 + 4t, \quad z = 4 + 2t$$

(b)

$$L_1 : x + 1 = 4t, \quad y - 3 = t, \quad z - 1 = 0$$

$$L_2 : x + 13 = 12t, \quad y - 1 = 6t, \quad z - 2 = 3t$$

20. Show that the lines L_1 and L_2 are skew.

(a)

$$L_1 : x = 1 + 7t, \quad y = 3 + t, \quad z = 5 - 3t$$

$$L_2 : x = 4 - t, \quad y = 6, \quad z = 7 + 2t$$

(b)

$$L_1 : x = 2 + 8t, \quad y = 6 - 8t, \quad z = 10t$$

$$L_2 : x = 3 + 8t, \quad y = 5 - 3t, \quad z = 6 + t$$

21. Determine whether the lines L_1 and L_2 are parallel.

(a)

$$L_1 : x = 3 - 2t, \quad y = 4 + t, \quad z = 6 - t$$

$$L_2 : x = 5 - 4t, \quad y = -2 + 2t, \quad z = 7 - 2t$$

(b)

$$L_1 : x = 5 + 3t, \quad y = 4 - 2t, \quad z = -2 + 3t$$

$$L_2 : x = -1 + 9t, \quad y = 5 - 6t, \quad z = 3 + 8t$$

Chapter: 11.6

22. Find an equation of the plane that passes through the point P and has the vector $\mathbf{n}$ as a normal.

(a) $P(2, 6, 1), \quad \mathbf{n} = \langle 1, 4, 2 \rangle$

(b) $P(-1, -1, 2), \quad \mathbf{n} = \langle -1, 7, 6 \rangle$

(c) $P(1, 0, 0), \quad \mathbf{n} = \langle 0, 0, 1 \rangle$

23. Find the distance between the point and the plane.

(a) $(1, -2, 3); \quad 2x - 2y + z = 4$

(b) $(0, 1, 5); \quad 3x + 6y - 2z - 5 = 0$

24. Find the distance between the given parallel planes.

(a) $-2x + y + z = 0, \quad 6x - 3y - 3z - 5 = 0$

(b) $x + y + z = 1, \quad x + y + z = -1$

25. Find the distance between the given skew lines.

(a)

$$x = 1 + 7t, \quad y = 3 + t, \quad z = 5 - 3t$$

$$x = 4 - t, \quad y = 6, \quad z = 7 + 2t$$

(b)

$$x = 3 - t, \quad y = 4 + 4t, \quad z = 1 + 2t$$

$$x = t, \quad y = 3 - t, \quad z = 2t$$

26. Find an equation of the plane that passes through the given points.

(a) $(-2, 1, 1), (0, 2, 3), (1, 0, -1)$

(b) $(3, 2, 1), (2, 1, -1), (-1, 3, 2)$

27. Determine whether the planes are parallel, perpendicular, or neither.

(a) $2x - 8y - 6z - 2 = 0, \quad -x + 4y + 3z - 5 = 0.$

(b) $x - y + 3z - 2 = 0, \quad 2x + z = 1$

(c) $3x - 2y + z = 4, \quad 6x - 4y + 3z = 7$

28. Determine whether the line and plane are parallel, perpendicular, or neither.

(a)

$$x = 4 + 2t, \quad y = -t, \quad z = -1 - 4t$$

$$3x + 2y + z - 7 = 0$$

(b)

$$x = 1, \quad y = 2t, \quad z = 3t$$

$$x - y + 2z = 5$$

(c)

$$x = -1 + 2t, \quad y = 4 + t, \quad z = 1 - r$$

$$4x + 2y - 2z = 7$$

29. (a)

$$x = 3 - r, \quad y = 2 + r, \quad z = 1 - 3r$$

$$2x + 2y - 5 = 0$$

(b)

$$x = 1 - 2r, \quad y = r, \quad z = -r$$

$$6x - 3y + 3z = 1$$

(c)

$$x = r, \quad y = 1 - t, \quad z = 2 + t$$

$$x + y + z = 1$$

30. Determine whether the line and plane intersect; if so, find the coordinates of the intersection.

(a)

$$x = t, \quad y = t, \quad z = t$$

$$3x - 2y + z - 5 = 0$$

(b)

$$x = 2 - t, \quad y = 3 + t, \quad z = t$$

$$2x + y + z = 1$$

(c)

$$x = 3t, \quad y = 5t, \quad z = -t$$

$$2x - y + z + 1 = 0$$

31. Find the acute angle of intersection of the planes to the nearest degree.

(a) $x = 0, \quad 2x - y + z - 4 = 0$

(b) $x + 2y - 2z = 5, \quad 6x - 3y + 2z = 8$

Chapter: 13.6

32. Find a unit vector in the direction in which f increases most rapidly at P , and find the rate of change of f at P in that direction.

$$f(x, y) = 4x^3y^2; \quad P(-1, 1)$$

33. Find a unit vector in the direction in which f decreases most rapidly at P and find the rate of change of f at P in that direction.

$$f(x, y) = 20 - x^2 - y^2; \quad P(-1, -3)$$

Chapter: 14.1

34. Evaluate the double integral:

$$\iint_R (5 - \cos 3x) \, dA$$

over the rectangle $R = \{(x, y) : 0 \leq x \leq \frac{\pi}{6}, \, 0 \leq y \leq 3\}$.

35. Evaluate the double integral:

$$\iint_R f(x, y) \, dA$$

over the rectangle $R = [0, 1] \times [-2, 5]$, where $f(x, y) = \frac{1}{x^2 + 1}$.

36. Evaluate the double integral:

$$\int_0^\pi \int_0^\pi x^2 y \sin(x^3 y) \, dx \, dy$$

37. Evaluate the iterated integral:

$$\int_0^1 \int_0^1 \frac{xy}{\sqrt{x^2 + y^2 + 1}} \, dx \, dy$$

Chapter: 14.2

38. Evaluate the double integral:

$$\int_0^1 \int_y^1 e^{-x^2} \, dx \, dy$$

39. Using a double integral to find the area enclosed by the equations:

$$y = -x, \quad -x + y = 2 \quad \text{and} \quad y = 2.$$

40. Use a double integral to find the volume of the region bounded above by the plane $x + y + z = 3$ and below by the rectangle with equations $x = -2$, $y = 1$, $x = 0$, and $y = 5$.

41. Use a double integral to find the area of the region R enclosed by:

$$y = -x + 1, \quad y = e^x, \quad y = 0, \quad \text{and} \quad x = 2$$

42. Find the volume of the solid bounded by $f(x, y) = 2xy$ and bounded region R in the xy -plane, where R is the region in the first quadrant enclosed between $y^2 = 4x$ and the line $y = \frac{1}{2}x$.

Chapter: 14.5

43. Evaluate the triple integral:

$$\int_0^2 \int_{-1}^1 \int_0^2 yz \, dx \, dz \, dy$$

44. Evaluate the triple integral:

$$\int_0^{\pi/4} \int_0^{x^2} \int_0^2 x \cos(y) \, dz \, dy \, dx$$

45. Evaluate the triple integral:

$$\iiint_G xy \sin(yz) \, dV$$

where G is the rectangular box defined by the inequalities $0 \leq x \leq \pi$, $0 \leq y \leq 1$, and $0 \leq z \leq \pi/6$.

46. Evaluate the triple integral:

$$\iiint_G y \, dV$$

where G is the solid enclosed by the plane $z = y$, the xy -plane, and the parabolic cylinder $y = 1 - x^2$.

47. Evaluate the triple integral:

$$\iiint_G xyz \, dV$$

where G is the solid in the first octant that is bounded by the parabolic cylinder $z = 2 - x^2$ and the planes $z = 0$, $y = x$, and $y = 0$.

48. Evaluate the triple integral:

$$\iiint_G \cos(z/y) \, dV$$

where G is the solid defined by the inequalities $\pi/6 \leq y \leq \pi/2$, $y \leq x \leq \pi/2$, and $0 \leq z \leq xy$.